



Fundamentos de Bases de Datos

Manual de Usuario para proyecto final

Carrera: Ciencias de la Computación

Asesor: Jose Saul De Lira Miramontes

Alumnos: Ivan Eduardo Urrutia Rodriguez 381396

Leonardo Franco Bulkey 377288

Fecha de entrega: 26/05/2026

Duo Builders Construction Management System

Illustrated User Manual

1. Purpose

The Duo Builders Construction Management System is a desktop application for running common construction company data queries from one organized interface. It lets users search construction sites, customers, employees, payments, payroll information, machines, materials, warehouses, and external services without typing sql commands.

2. Starting the app

To open the app, double click: `run_construction_manager.bat`

You can also start it from powershell with: `python construction_manager_app.py`

If Python is installed but the Oracle package is missing, install it with: `pip install oracledb`

3. Login

When the app opens, enter the login credentials:

User: `ConstructionCompany`

Password: `const123`

Click Login or press Enter.

If the login is incorrect, the app clears the password field and shows an invalid login message.



The login screen for the Duo Builders Construction Management System features a logo at the top consisting of three stylized buildings in blue and green, with a blue swoosh underneath. Below the logo, the text "Duo Builders" is displayed in a large, bold, blue font. Underneath that, "Construction Management System" is written in a smaller, bold, black font. A subtitle in a small, regular black font reads "Sign in to manage projects, payroll, materials, and services." The login form includes a "User" label above a text input field, a "Password" label above another text input field, and a blue "Login" button at the bottom.

Figure 1. Login screen for the Duo Builders Construction Management System.

4. Main Screen Overview

After logging in, the main screen shows three main areas:

- **Modules:** The left sidebar where you choose the area of the company database.
- **Query:** The top panel where you choose a specific query and enter any required fields.
- **Results:** The bottom panel where query results, messages, or errors are displayed.

The status line at the bottom shows whether the app is ready, running a query, done, or if a query failed.

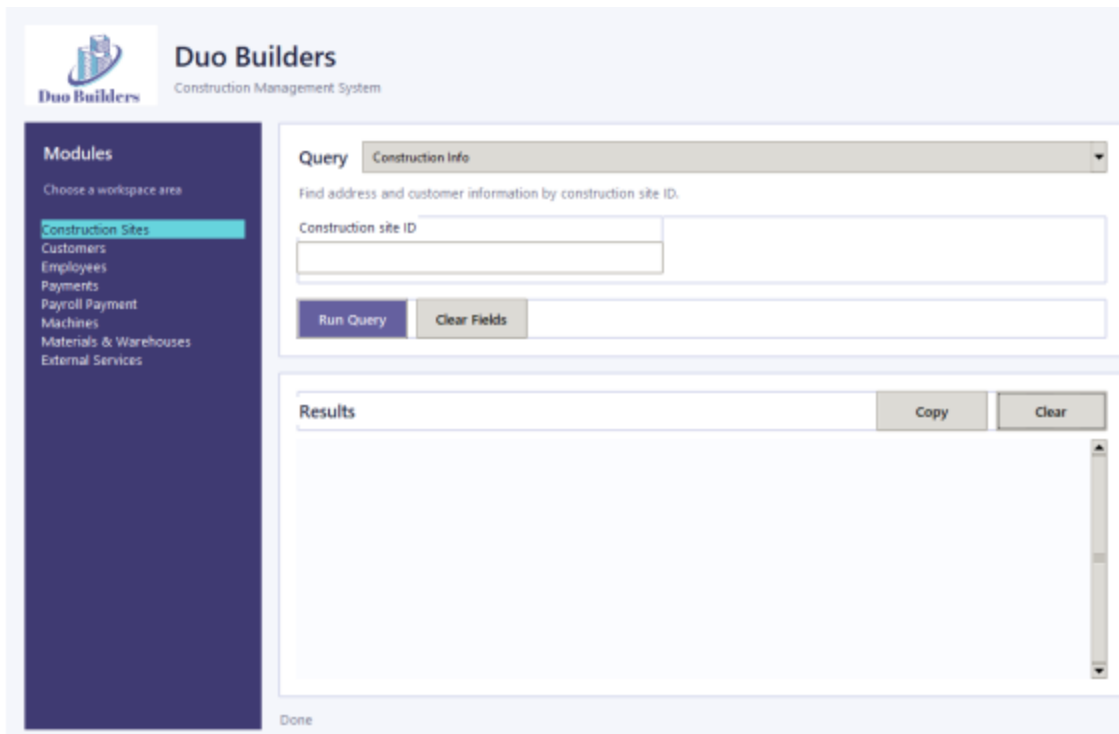


Figure 2. Main application screen after logging in.

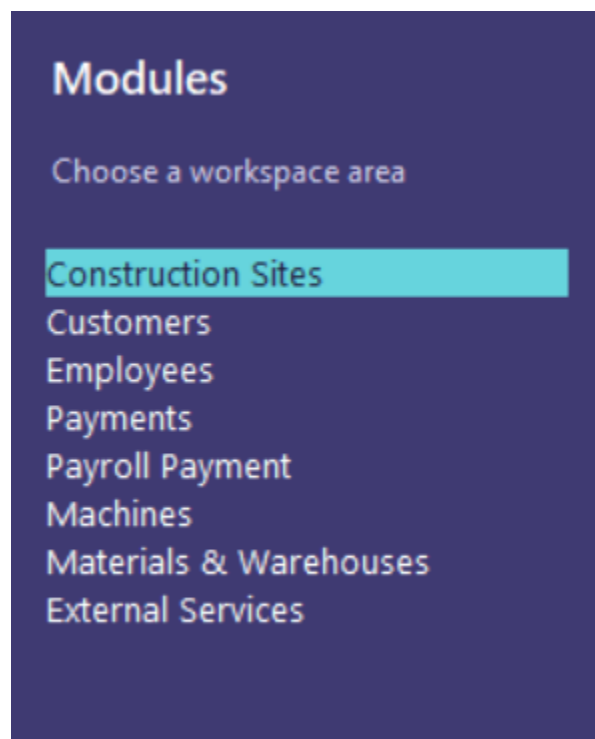


Figure 3. Modules sidebar used to choose the database area

5. Running A Query

- Select a module from the left sidebar
- Select a query from the Query dropdown
- Fill in the required fields, if any appear.
- Click Run Query
- Read the output in the Results panel.

The app runs queries in the background, so the window should stay responsive while Oracle is working.

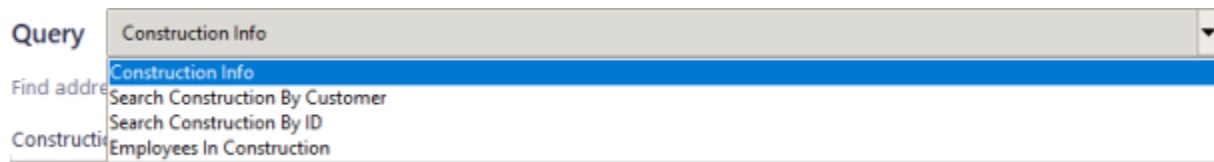


Figure 4. Query dropdown showing the available queries for the selected module

A screenshot of a web application interface showing a query form and its results. The "Query" dropdown is set to "Employee Info". Below it, the instruction "Find contact and employment dates by employee ID." is displayed. There is a text input field labeled "Employee ID" containing the value "E001". Below the input field are two buttons: "Run Query" and "Clear Fields". Below the form, there is a "Results" section with a "Copy" button and a "Clear" button. The results area contains the text: "Carlos Lopez Start date: 2025-01-20 Hire date: 2024-08-01 Phone: 6143000001". At the bottom left, the word "None" is visible.

Figure 5. Example query using an employee ID and displaying the result.

6. Clearing Fields And Results

Use Clear Fields to erase the input boxes for the selected query.

Use Clear in the Results panel to erase the displayed output.

Use Copy in the Results panel to copy the current results to the clipboard.



Figure 6. Results panel actions for copying or clearing displayed output.

7. Modules And Queries

Construction Sites

Use this module to search for construction site information.

Available queries:

- Construction Info: Enter a construction site ID to find address and customer information.
- Search Construction By Customer: Enter a customer name to search construction sites.
- Search Construction By ID: Enter a construction site ID to find a specific site.
- Employees In Construction: Enter a construction site ID to list assigned employees.

Customers

Use this module to look up customer information.

Available queries:

- Customer Info: Enter a customer ID to find phone, email, and customer type.
- Customer Type: Enter a customer ID to find the customer's type.

Employees

Use this module to search employee information.

Available queries:

- Employees Under Managers: Lists employees together with their manager.

- Employee Info: Enter an employee ID to find contact and employment dates.
- Employees By Bank: Enter a bank name to list employees who use that bank.
- Employees By License: Enter a license type to list employees with that license.

Payments

Use this module to review employee payment records.

Available queries:

- Paid Amount: Enter year, week number, and employee ID to check the paid amount.
- Deductible Tax: Enter year, week number, and employee ID to check tax.
- Week Was Paid: Enter year, week number, and employee ID to check whether the week was paid.

Payroll Payment

Use this module to review and register payroll payments.

Available queries:

- Employees To Pay: Enter year and week number to list employees with payment information for that week.
- Register Employee Payment: Enter year, week number, and employee ID to register one employee payment.

When Register Employee Payment is selected, the app automatically loads an employee ID/name list in the Results panel to help choose the correct employee.

Results

Employees:

E001		Carlos Lopez
E002		Maria Garcia
E003		Jose Hernandez
E004		Lucia Ramos
E005		Miguel Torres
E006		Sofia Vega
E007		Raul Mendoza
E008		Paola Castro
E009		Diego Salazar
E010		Elena Morales

Figure 7. Employee list shown when registering an employee payment.

If the employee was already paid for the same year and week, the app displays:

This employee has already been paid.

Query

Register Employee Payment

Pay one employee for a week using CurrentPay and the employee's last tax value.

Year	Week number	Employee ID
1	2	E001

Run Query

Clear Fields

Results

Copy

Clear

Payment registered successfully.
Employee ID: E001
Employee: Carlos Lopez
Week: 2
Year: 1
Current pay: 14500.00
Payment amount: 3625.00
Tax deducted: 420.00
Net payment: 3205.00

Figure 8. Register Employee Payment query after a successful payment.

Machines

Use this module to view machine information.

Available queries:

- All Machines: Lists all machines and whether each one is in use.
- Machines In Construction: Enter a construction site ID to list machines assigned to that site.

Materials & Warehouses

Use this module to search material, stock, and warehouse information.

Available queries:

- Material Purpose: Enter a material name to find its purpose.
- Current Stock Quantity: Enter a warehouse ID and material ID to find stock quantity.
- All Warehouses: Lists all warehouses.
- Warehouses In Construction: Enter a construction site ID to list warehouses assigned to that site.

External Services

Use this module to search external service company information.

Available queries:

- External Companies: Lists external service companies.
- Companies By Service Type: Enter a service type to list companies that provide that service.

8. Input Rules

All visible input fields are required before running a query.

Fields such as Year and Week number must be whole numbers. If a number field contains letters, symbols, or decimals, the app shows a warning.

Text fields such as customer name, bank name, license type, material name, employee ID, and construction site ID should match the values stored in the database.

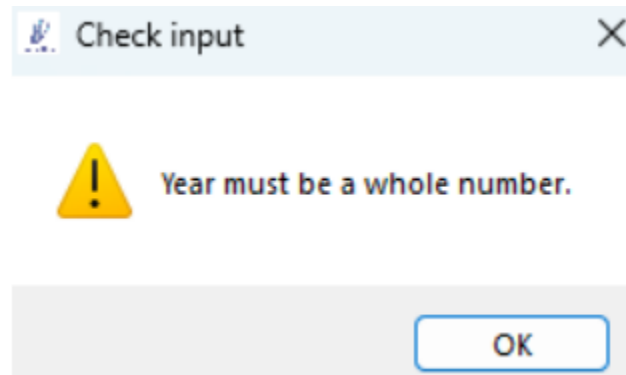


Figure 9. Warning shown when a number field does not contain a whole number.

9. Results And Error Messages

Successful queries display their output in the Results panel.

If Oracle or the backend script returns an error, the app displays the error message in the Results panel and changes the status to Query failed. Common causes include:

- Oracle Database is not running.
- The oracledb Python package is not installed.
- The database user does not exist or is locked.
- The backend script username, password, or DSN does not match the local Oracle setup.
- The entered ID or name does not match data in the database.

10. Oracle Login Troubleshooting

The app login uses:

User: ConstructionCompany

Password: const123

The backend scripts also expect Oracle to use:

User: ConstructionCompany

Password: const123

DSN: localhost:1521/XEPDB1

If the app login works but a query shows ORA-01017, Oracle rejected the database credentials. Create or unlock the Oracle user with the same password, or update the backend script connection constants to match your Oracle user.

11. Closing the app

To close the application, click the X button in the window title bar.

Any query results shown in the Results panel are not saved automatically. Use Copy if you need to keep the output.